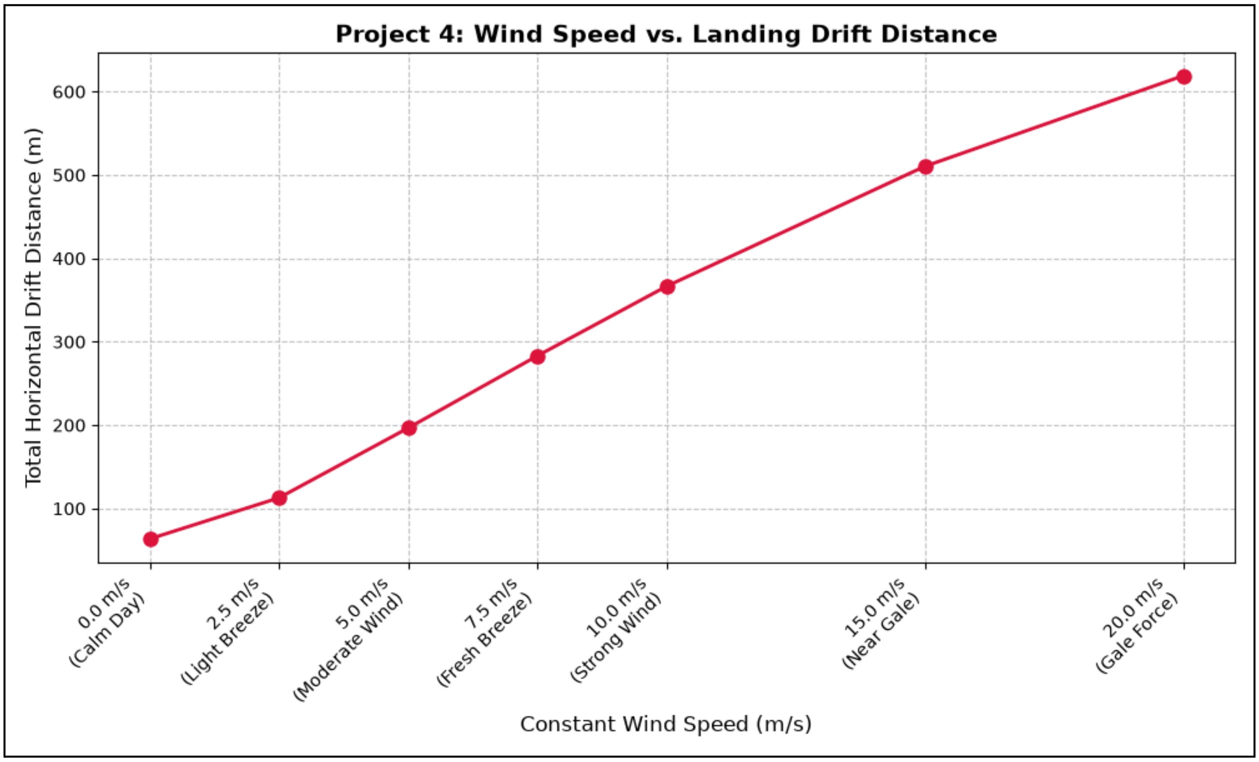
figure 1a



- the upward-sloping red line illustrates how higher wind speeds directly increase the distance the rocket is blown off course during its parachute descent
- moving from a "Calm Day" (0.0 m/s wind) to "Gale Force" (20.0 m/s wind), the horizontal drift expands dramatically, pushing the final landing zone over 600 meters away from the launch pad

output

Condition	Wind (m/s)	Max Altitude (m)	Drift Dist. (m)	Landing (X, Y)
Calm Day	0.0	1708.63	63.55	(-0.2, 63.6)
Light Breeze	2.5	1707.56	112.77	(93.6, 62.9)
Moderate Wind	5.0	1704.30	196.53	(186.7, 61.4)
Fresh Breeze	7.5	1698.89	283.04	(276.7, 59.7)
Strong Wind	10.0	1691.51	366.45	(361.9, 57.7)
Near Gale	15.0	1671.20	510.33	(507.6, 53.1)
Gale Force	20.0	1646.12	618.96	(617.1, 47.9)

- the table shows the physical landing spot (X, Y) shifting further away as the total drift jumps from 63.55 meters to 618.96 meters
- the data reveals that flying into stronger winds actually reduces the rocket's maximum altitude, causing it to drop from a peak height of 1708.63 meters down to 1646.12 meters